Insert new Clause 10 after Clause 9 as follows:

10. YANG data model

This clause specifies the YANG (IETF RFC 7950) data model that provides control and status monitoring of systems and system components that implement functionality specified in this standard.

This clause:

a) Introduces the YANG framework that governs the naming and hierarchy of configuration and operational data structures in the data models, and the modeling of network interfaces (10.1).
b) Describes the information data model and its relationship to the operational processes and managed objects specified in the other clauses of this standard, and provides a representation, similar to the Unified Modeling Language (UML), of relevant objects (10.2).
c) Describes the structure of the YANG data model, which comprises three YANG modules (10.3).
d) Includes a relationship description of other modules imported in YANG modules (10.4)
e) Reviews security considerations applicable to each of the modules, with specific reference to data nodes in the YANG modules that compose the YANG data model (10.5).
f) Includes each of the YANG modules (10.7) and its data schema (10.6).

10.1 YANG framework

The YANG framework applies hierarchy in the following areas:

a) The uniform resource name (URN), as specified in IEEE Std 802 for the IEEE 802 series of standards.
b) The YANG objects form a hierarchy of configuration and operational data structures that specify the YANG data model.

10.2 Information model for Link Aggregation management

The YANG objects are based on the managed objects in Clause 7. A UML-like representation of the management model is provided in the following subclauses.

NOTE—OMG® UML2.5[9] [B11] conventions together with C++ language constructs are used in this clause as a representation to convey model structure and relationships.

The purpose of a UML-like[10] diagram is to express the model design on a single piece of paper. The structure of the UML-like representation shows the name of the object followed by a list of properties for the object. The properties indicate its type and accessibility. The UML-like representation is meant to express simplified semantics for the properties. It is not meant to provide the specific datatype as used to encode the object in either MIB or YANG. In the UML-like representation, a box with a white background represents information that comes from sources outside of the IEEE. A box with a gray background represents objects that are specified by this IEEE Standard.

The YANG hierarchical structure that incorporates the Link Aggregation YANG modules supported by this standard is represented by Figure 10-1. In the figures in this clause, items that are shaded gray are described in this document, items with no background shading are specified elsewhere. The YANG data model is

[9] OMG® is a registered trademark of the Object Management Group, Inc.
[10] A description of the UML-like diagrams used in this clause is provided at https://1.ieee802.org/uml-like-diagrams.

realized in three YANG modules. One module *ieee802-dot1ax-types* provides data types that are needed by the Link Aggregation configuration and monitoring objects. The *ieee802-dot1ax-linkagg* module provides the Link Aggregation, Aggregation Port, and Aggregator configuration and monitoring objects. The *ieee802-dot1ax-drni* module provides the Distributed Resilient Network Interface (DRNI) configuration and monitoring objects. The Link Aggregation and DRNI capabilities are not only applicable to IEEE Std 802.1Q Bridges, but also (for example) end stations and routers.

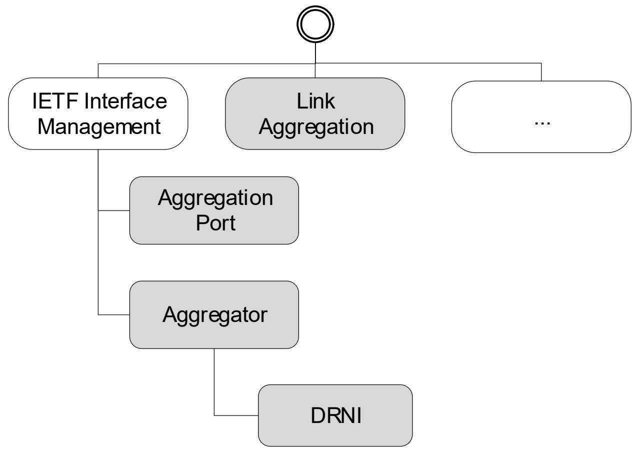

Figure 10-1—YANG root hierarchy with Link Aggregation YANG modules

10.2.1 Link Aggregation model

The Link Aggregation configuration and monitoring objects in Figure 10-2 show the objects that are applicable on a system supporting Link Aggregation, and the interfaces supporting Link Aggregation.

The objects on the system consist of a list of Aggregation Systems, and a list of key groups. A key group is the set of Aggregators and Aggregation Ports that share the same System Priority, System MAC Address, and Aggregation Key, and therefore can potentially form a Link Aggregation Group. Each key group has a unique combination of actor-admin-key and actor-system, and includes parameters that have the same value for any Aggregator and/or Aggregation Port that have the same actor-admin-key and actor-system.

NOTE—See 6.3.5 and 6.3.6 for a description of the identification of Aggregation Ports that can potentially form a Link Aggregation Group, and Figure 6-15 for examples.

Each Aggregation Port is an interface (e.g., of type ethernetCsmacd) augmented with an “aggport” presence container as shown on the right side of Figure 10-2. Each Aggregator is an interface to the entire Link Aggregation Group (using type ieee8023adlag[11]) augmented with a “lag” container as shown on the left side of Figure 10-2. The binding of Aggregation Ports to Aggregators is a dynamic function of the Link Aggregation Control Protocol. The Aggregation Ports attached to an Aggregator can be read from the *lower-layer-if* attribute of the Aggregator interface. The Aggregator to which an Aggregation Port is attached can be read through the *higher-layer-if* attribute of the Aggregation Port interface. An Aggregation Port only transmits and responds to LACPDUs when the interface “enabled” leaf is true, and only attaches to an Aggregator when the Aggregator interface “enabled” leaf is true. The interface “oper-status” leaf of an Aggregation Port is “up” when the Aggregation Port is attached to an Aggregator and the “collecting” and “distributing” bits of the “actor-oper-state” leaf are both true. The interface “oper-status” leaf of an Aggregator is “up” when it has at least one attached Aggregation Port that is “up”.

[11] This type reflects the fact that the Link Aggregation specification was originally standardized in IEEE Std 802.3ad-2000. Further development has taken place in the IEEE 802.1 Working Group since 2008. Subsequent revisions have been published as IEEE Std 802.1AX.

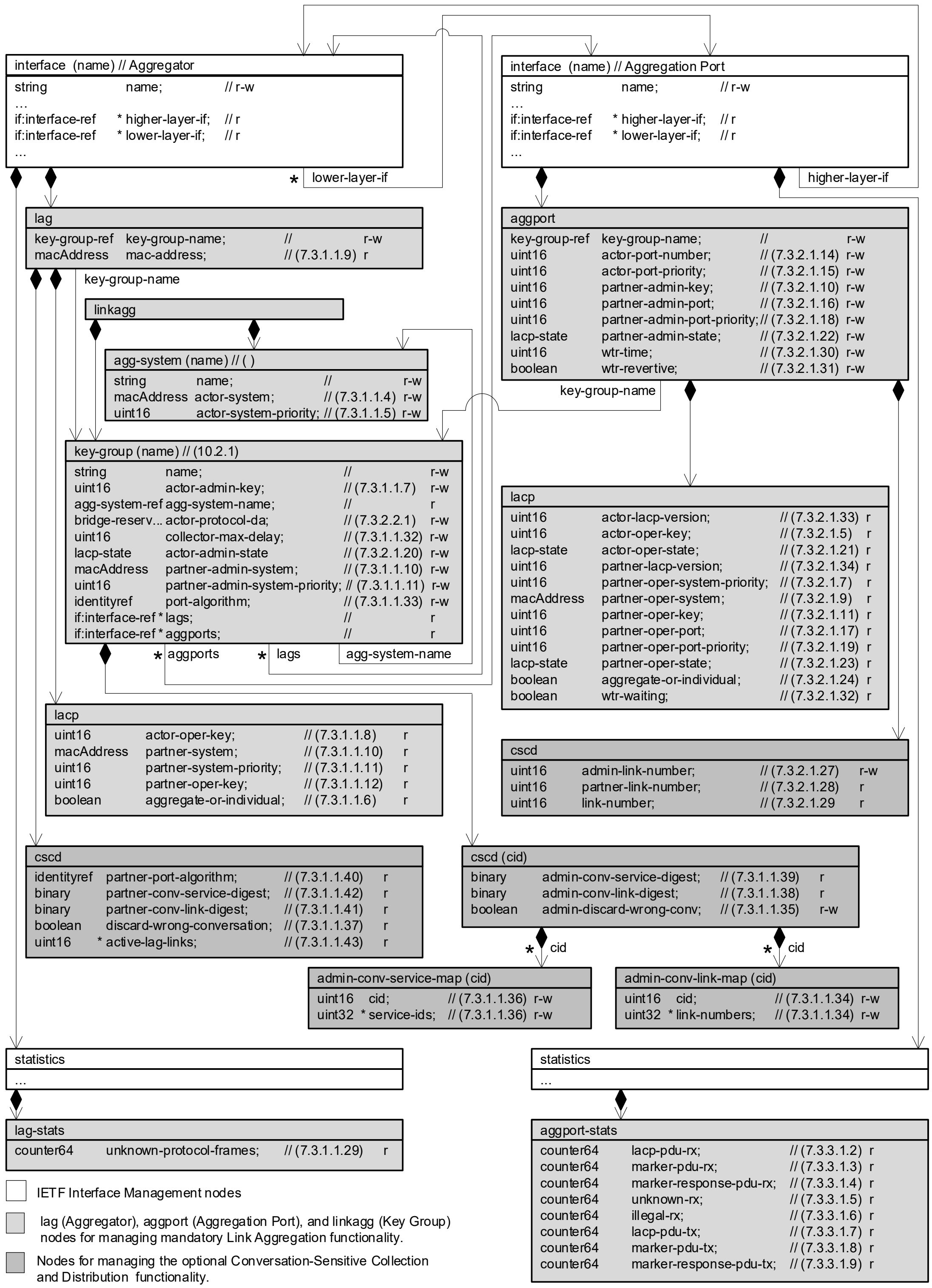

Figure 10-2—Link Aggregation configuration and monitoring objects

10.2.2 DRNI model

The DRNI configuration and monitoring objects in Figure 10-3 show the objects that are applicable to an Aggregator augmented with a "drni" container.

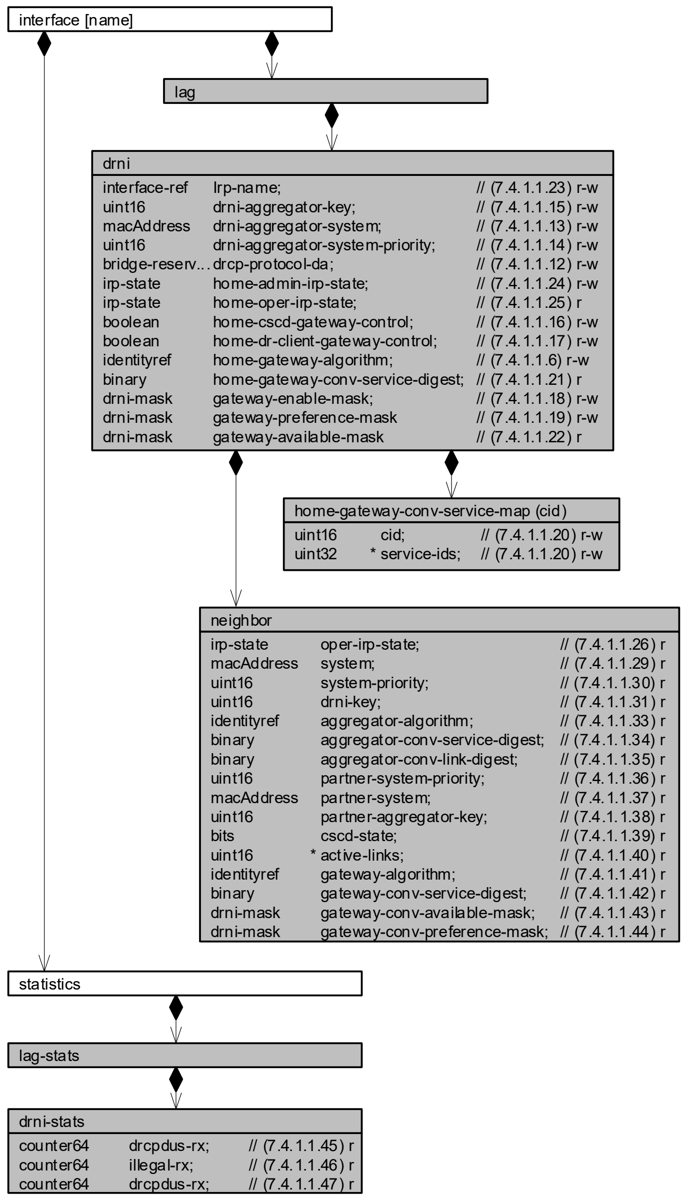

Figure 10-3—DRNI configuration and monitoring objects

10.3 Structure of the Link Aggregation YANG data model

The YANG data model specified in this standard is composed of three YANG modules. A summary of the modules contained in this clause is represented in Table 10-1.

Table 10-1—Structure of the YANG modules

Module	Subclause	Notes
ieee802-dot1ax-types	10.7.1	Type definitions used for Link Aggregation YANG.
ieee802-dot1ax-linkagg	10.7.2	Link Aggregation Management
ieee802-dot1ax-drni	10.7.3	DRNI Management

In the YANG modules below, if any discrepancy between the DESCRIPTION text and the corresponding specification in any other part of this standard occurs, the specifications outside this clause take precedence.

10.4 Relationship to other YANG modules

This clause describes how the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules are related to the YANG modules that are imported.

10.4.1 IEEE 802.1AX Types module

The *ieee802-dot1ax-types* module provides reusable types that are used by the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* modules.

10.4.2 IETF YANG Types module

The *ietf-yang-types* YANG module (IETF RFC 6991) contains a set of derived YANG types. This document leverages counter64.

10.4.3 IETF Interfaces YANG module

The *ietf-interfaces* YANG module (IETF RFC 8343) contains a set of YANG definitions for managing network interfaces. This document augments an ietf-interfaces:interface with aggregation-port or aggregator data nodes.

10.4.4 IEEE 802 Types module

The *ieee802-types* module provides reusable types that are used in IEEE 802 standards.

NOTE—The type for mac-addresses defined in *ieee802-types* has a pattern that allows upper and lower case letters. To avoid issues with string comparison, it is suggested to only use upper case for the letters in the hexadecimal numbers. Implementers using code comparing MAC addresses are advised that there is still an issue with a difference between the IETF mac-address definition and the IEEE mac-address definition.

10.5 Security considerations

The YANG modules specified in this clause are designed to be accessed via a network configuration protocol (e.g., the NETCONF protocol). In the case of NETCONF, the lowest NETCONF layer is the secure transport layer and the mandatory to implement secure transport is SSH. The NETCONF access control model provides the means to restrict access for particular NETCONF users to a pre-configured subset of all available NETCONF protocol operations and content.

It is the responsibility of a system's implementor and administrator to ensure that the protocol entities in the system that support NETCONF, and any other remote configuration protocols that make use of these YANG modules, are properly configured to allow access only to those users who have legitimate rights to read or write data nodes. This standard does not specify how the credentials of those users are to be stored or validated.

10.5.1 Security considerations of the *ieee802-dot1ax* YANG modules

There are several management objects specified in the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules that are configurable (i.e., read-write) and/or operational (i.e., read-only). Such objects could be considered sensitive or vulnerable in some network environments. A network configuration protocol, such as NETCONF (IETF RFC 6241), can support protocol operations that can edit or delete YANG module configuration data (e.g., edit-config, delete-config, copy-config). If this is done in a non-secure environment without proper protection, then negative effects on the network operation are possible.

The following containers, and the objects in these containers, of the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules can be manipulated to interfere with the operation of the Link Aggregation Control Protocol (LACP) or the Distributed Relay Control Protocol (DRCP). This could, for example, cause network instability and result in the loss of service for a large number of end users.

dot1ax-linkagg/linkagg/key-group
dot1ax-linkagg/linkagg/key-group/cscd
dot1ax-linkagg/linkagg/key-group/cscd/admin-conv-service-map
dot1ax-linkagg/linkagg/key-group/cscd/admin-conv-link-map
dot1ax-linkagg/aggport
dot1ax-linkagg/aggport/cscd/admin-link-number
dot1ax-linkagg/lag
dot1ax-drni/lag/drni
dot1ax-drni/lag/drni/home-gateway-conv-service-map

Some of the readable data in this YANG module could be considered sensitive or vulnerable in some network environments. It is important to control all types of access (e.g., including NETCONF get, get-config operations) to these objects and possibly to even encrypt the values of these objects when sending them over the network. For example, the system name and other information about the remote systems could provide information about the configuration and topology of the network and could be considered a privacy threat.

10.6 YANG schema tree definitions

A simplified graphical representation of the data model is used in this document. The meaning of the symbols in these diagrams is as follows:

— Brackets "[" and "]" enclose list keys.

— Abbreviations before data node names: “rw” means configuration (read-write) and “ro” means state data (read-only).
— Symbols after data node names: “?” means an optional node, “!” means a presence container, and “*” denotes a list and leaf-list.
— Parentheses enclose choice and case nodes, and case nodes are also marked with a colon (“:”).
— Ellipses (“…”) stand for contents of subtrees that are not shown.

10.6.1 Schema for the *ieee802-dot1ax-linkagg* YANG module

```
module: ieee802-dot1ax-linkagg
  +--rw linkagg
     +--rw agg-system* [name]
     |  +--rw name                     string
     |  +--rw actor-system?            ieee:mac-address
     |  +--rw actor-system-priority?   uint16
     +--rw key-group* [name]
        +--rw name                             string
        +--rw actor-admin-key                  uint16
        +--rw agg-system-name                  agg-system-ref
        +--rw actor-protocol-da?               ax:bridge-reserved-addresses
        +--rw collector-max-delay?             uint16
        +--rw actor-admin-state?               ax:lacp-state
        +--rw partner-admin-system?            ieee:mac-address
        +--rw partner-admin-system-priority?   uint16
        +--rw port-algorithm?                  identityref
        +--ro lags*                            if:interface-ref
        +--ro aggports*                        if:interface-ref
        +--rw cscd {ax:cscd}?
           +--rw admin-conv-service-map {ax:sid-map}?
           |  +--rw (method)?
           |     +--:(pattern)
           |     |  +--rw pattern?    identityref
           |     +--:(cid-list)
           |        +--rw cid-list* [cid]
           |           +--rw cid            uint16
           |           +--rw service-ids*   uint32
           +--ro admin-conv-service-digest?   binary
           +--rw admin-conv-link-map
           |  +--rw (method)?
           |     +--:(pattern)
           |     |  +--rw pattern?    identityref
           |     +--:(cid-list)
           |        +--rw cid-list* [cid]
           |           +--rw cid             uint16
           |           +--rw link-numbers*   uint16
           +--ro admin-conv-link-digest?      binary
           +--rw admin-discard-wrong-conv?    enumeration {ax:dwc}?

  augment /if:interfaces/if:interface:
    +--rw lag
       +--rw key-group-name    key-group-ref
       +--ro mac-address?      ieee:mac-address
       +--ro lacp
       |  +--ro actor-oper-key?            uint16
       |  +--ro partner-system?            ieee:mac-address
       |  +--ro partner-system-priority?   uint16
       |  +--ro partner-oper-key?          uint16
       |  +--ro aggregate-or-individual?   boolean
       +--ro cscd {ax:cscd}?
          +--ro partner-port-algorithm?        identityref
          +--ro partner-conv-service-digest?   binary
          +--ro partner-conv-link-digest?      binary
          +--ro discard-wrong-conversation?    boolean
          +--ro active-lag-links*              uint16
  augment /if:interfaces/if:interface/if:statistics:
    +--ro lag-stats
       +--ro unknown-protocol-frames?   yang:counter64
  augment /if:interfaces/if:interface:
    +--rw aggport!
```

```
      +--rw key-group-name                 key-group-ref
      +--rw actor-port-number?             uint16
      +--rw actor-port-priority?           uint16
      +--rw partner-admin-key?             uint16
      +--rw partner-admin-port?            uint16
      +--rw partner-admin-port-priority?   uint16
      +--rw partner-admin-state?           ax:lacp-state
      +--rw wtr-time?                      uint16
      +--rw wtr-revertive?                 boolean
      +--ro lacp
      |  +--ro actor-lacp-version?             uint8
      |  +--ro actor-oper-key?                 uint16
      |  +--ro actor-oper-state?               ax:lacp-state
      |  +--ro partner-lacp-version?           uint8
      |  +--ro partner-oper-system-priority?   uint16
      |  +--ro partner-oper-system?            ieee:mac-address
      |  +--ro partner-oper-key?               uint16
      |  +--ro partner-oper-port?              uint16
      |  +--ro partner-oper-port-priority?     uint16
      |  +--ro partner-oper-state?             ax:lacp-state
      |  +--ro aggregate-or-individual?        boolean
      |  +--ro wtr-waiting?                    boolean
      +--rw cscd {ax:cscd}?
         +--rw admin-link-number?     uint16
         +--ro partner-link-number?   uint16
         +--ro link-number?           uint16
  augment /if:interfaces/if:interface/if:statistics:
    +--ro aggport-stats
      +--ro lacp-pdu-rx?              yang:counter64
      +--ro marker-pdu-rx?            yang:counter64
      +--ro marker-response-pdu-rx?   yang:counter64
      +--ro unknown-rx?               yang:counter64
      +--ro illegal-rx?               yang:counter64
      +--ro lacp-pdu-tx?              yang:counter64
      +--ro marker-pdu-tx?            yang:counter64
      +--ro marker-response-pdu-tx?   yang:counter64
```

10.6.2 Schema for the *ieee802-dot1ax-drni* YANG module

```
module: ieee802-dot1ax-drni

  augment /if:interfaces/if:interface/dot1ax:lag:
    +--rw drni!
      +--rw irp-name                           if:interface-ref
      +--rw drni-aggregator-key?               uint16
      +--rw drni-aggregator-system?            ieee:mac-address
      +--rw drni-aggregator-system-priority?   uint16
      +--rw drcp-protocol-da?                  ax:bridge-reserved-addresses
      +--rw home-admin-irp-state?              irp-state
      +--ro home-oper-irp-state?               irp-state
      +--rw home-cscd-gateway-control?         boolean
      +--rw home-dr-client-gateway-control?    boolean
      +--rw home-gateway-algorithm?            identityref
      +--rw home-gateway-conv-service-map {ax:sid-map}?
      |  +--rw (method)?
      |     +--:(pattern)
      |     |  +--rw pattern?    identityref
      |     +--:(cid-list)
      |        +--rw cid-list* [cid]
      |           +--rw cid            uint16
      |           +--rw service-ids*   uint32
      +--ro home-gateway-conv-service-digest?   binary
      +--rw gateway-enable-mask
      |  +--rw cid-str*       string
      |  +--rw invert-mask?   boolean
      +--rw gateway-preference-mask
      |  +--rw cid-str*       string
      |  +--rw invert-mask?   boolean
      +--ro gateway-available-mask
      |  +--ro cid-str*       string
```

```
       |  +--ro invert-mask?   boolean
       +--ro neighbor
          +--ro oper-irp-state?                  irp-state
          +--ro system?                          ieee:mac-address
          +--ro system-priority?                 uint16
          +--ro drni-key?                        uint16
          +--ro aggregator-algorithm?            identityref
          +--ro aggregator-conv-service-digest?  binary
          +--ro aggregator-conv-link-digest?     binary
          +--ro partner-system-priority?         uint16
          +--ro partner-system?                  ieee:mac-address
          +--ro partner-aggregator-key?          uint16
          +--ro cscd-state?                      bits
          +--ro active-links*                    uint16
          +--ro gateway-algorithm?               identityref
          +--ro gateway-conv-service-digest?     binary
          +--ro gateway-available-mask
          |  +--ro cid-str*       string
          |  +--ro invert-mask?   boolean
          +--ro gateway-preference-mask
             +--ro cid-str*       string
             +--ro invert-mask?   boolean
  augment /if:interfaces/if:interface/if:statistics/dot1ax:lag-stats:
    +--ro drni-stats
       +--ro drcpdus-rx?   yang:counter64
       +--ro illegal-rx?   yang:counter64
       +--ro drcpdus-tx?   yang:counter64
```

10.7 YANG modules[12,13,14]

10.7.1 The *ieee802-dot1ax-types* YANG module

```
module ieee802-dot1ax-types {
  yang-version 1.1;
  namespace "urn:ieee:std:802.1AX:yang:ieee802-dot1ax-types";
  prefix dot1ax-types;

  organization
    "IEEE 802.1 Working Group";
  contact
    "WG-URL: http://www.ieee802.org/1/
     WG-Email: stds-802-1-l@ieee.org

     Contact: IEEE 802.1 Working Group Chair
      Postal: C/O IEEE 802.1 Working Group
              IEEE Standards Association
              45 Hoes Lane
              Piscataway, NJ  08854
              USA
      E-mail: stds-802-1-chairs@ieee.org";
  description
    "Common types used within 802.1AX Link Aggregation modules.

     Copyright (C) IEEE (2025).

     This version of this YANG module is part of IEEE Std 802.1AX;
     see the standard itself for full legal notices.";

  revision 2025-09-~~05 ~~30 {
    description
```

[12] *Copyright release for YANG modules*: Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

[13] An ASCII version of the YANG module(s) can be obtained by Web browser from the IEEE 802.1 Website at https://1.ieee802.org/yang-modules/.

[14] References in this standard's YANG module definitions are not clickable, as each module has been incorporated unchanged after development and verification using YANG tools.

```
      "Published as part of IEEE Std 802.1AXdz-2025.
       The following reference statement identifies each ~~referenced IEEE~~referenced
       IEEE Standard as updated by applicable amendments.";
    reference
      "IEEE Std 802.1AX-2020 Link Aggregation:
       IEEE Std 802.1AXdz-2025.";
  }

  feature cscd {
    description
      "~~Conversation ~~Conversation-Sensitive Collection and Distribution (CSCD)
       is supported.";
    reference
      "5.3.2, 6.6 of IEEE Std 802.1AX";
  }

  feature dwc {
    description
      "The Discard Wrong Conversation option in CSCD is
       supported.";
    reference
      "5.3.2, 6.6 of IEEE Std 802.1AX";
  }

  feature sid-map {
    description
      "The Service ID to Conversation ID map is supported for
       use by distribution algorithms.";
  }

  feature c-vid-dist-alg {
    description
      "Supports distribution based on C-VIDs.";
  }

  feature s-vid-dist-alg {
    description
      "Supports distribution based on S-VIDs.";
  }

  feature i-sid-dist-alg {
    description
      "Supports distribution based on I-SIDs.";
  }

  feature te-sid-dist-alg {
    description
      "Supports distribution based on TE-SIDs.";
  }

  feature flow-hash-dist-alg {
    description
      "Supports distribution based on Flow Hash.";
  }

  feature basic-link-maps {
    description
      "Pre-defined admin-conv-link-map configurations to distribute
       packets on either one or two active links, with any
       remaining links available as standby.";
  }

  identity distribution-algorithm {
    description
      "Each distribution algorithm is identified by a sequence of
       4 octets, structured as shown in Figure 8-1. Distribution
       algorithm identifiers are used by network administrators to
       select between algorithms and, in Conversation-sensitive
       LACP and (if supported) Distributed Resilient Network
       Interconnect (DRNI) operation, to check whether partners
       and neighbors are using the same algorithm.
```

```
       This identity is intended to serve as base identity, not
       to be directly referenced.

       Vendor specific, combination (ex: multi-layer), and other
       customized distribution algorithms can be created as
       their own identities in their own YANG files, derived from
       this imported base type.  When the identity description
       specifies a ~~4 ~~4-octet value, this value is used for ~~csed~~CSCD
       purposes, and included in LACPDUs. Otherwise the identity
       is used purely for local selection of a distribution
       algorithm, and the unspecified distribution algorithm is
       used for ~~csed ~~CSCD purposes.";
    reference
      "8.1, 8.2 of IEEE Std 802.1AX";
  }

  identity unspecified {
    base distribution-algorithm;
    description
      "The 'Unspecified distribution algorithm' identifier has
       been reserved for use when the algorithm is unknown (or
       is not advertised).
       The algorithm identifier is 00-80-C2-00.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity c-vids-nomap {
    if-feature "c-vid-dist-alg";
    base distribution-algorithm;
    description
      "Distribution based on C-VIDs (8.2.1). No Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-01.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity c-vids-map {
    if-feature "c-vid-dist-alg and sid-map";
    base distribution-algorithm;
    description
      "Distribution based on C-VIDs (8.2.1). A Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-81.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity s-vids-nomap {
    if-feature "s-vid-dist-alg";
    base distribution-algorithm;
    description
      "Distribution based on S-VIDs (8.2.2). No Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-02.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity s-vids-map {
    if-feature "s-vid-dist-alg and sid-map";
    base distribution-algorithm;
    description
      "Distribution based on S-VIDs (8.2.2). A Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-82.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity i-sids-nomap {
```

```
    if-feature "i-sid-dist-alg";
    base distribution-algorithm;
    description
      "Distribution based on I-SIDs (8.2.3). No Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-03.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity i-sids-map {
    if-feature "i-sid-dist-alg and sid-map";
    base distribution-algorithm;
    description
      "Distribution based on I-SIDs (8.2.3). A Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-83.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity te-sids-nomap {
    if-feature "te-sid-dist-alg";
    base distribution-algorithm;
    description
      "Distribution based on TE-SIDs (8.2.4). No Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-04.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity te-sids-map {
    if-feature "te-sid-dist-alg and sid-map";
    base distribution-algorithm;
    description
      "Distribution based on TE-SIDs (8.2.4). A Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-84.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity flow-hash-nomap {
    if-feature "flow-hash-dist-alg";
    base distribution-algorithm;
    description
      "Distribution based on Flow Hash (8.2.5). No Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-05.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity flow-hash-map {
    if-feature "flow-hash-dist-alg and sid-map";
    base distribution-algorithm;
    description
      "Distribution based on Flow Hash (8.2.5). A Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-85.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity link-map-patterns {
    description
      "Base identity for patterns filling the admin-conv-link-map.
       This identity is intended to serve as base identity, not
       to be directly referenced. Use of the identity allows vendors
       to augment the module with vendor specified patterns.";
    reference
```

```
      "7.3.1.1.34, 6.6.3.1 of IEEE Std 802.1AX";
  }

  identity one-plus-n {
    if-feature "basic-link-maps";
    base link-map-patterns;
    description
      "Provides active/standby behavior with one link carrying
       all traffic and any other links in the LAG are standby.
       All packets are mapped to the lowest link number that is
       active in the aggregation. This is updated whenever a link
       becomes active or inactive in the aggregation. The pattern
       supports any number of links with link numbers between 1
       and 4095. Any links in the aggregation with a link number
       greater than 4095 are not used.
       The map consists of an identical entry for each CID, with
       a list of numbers from 1 to 4095, followed by the number 0.
       The MD5 digest calculated for this map is
       0xd2dc4cf241740239408945bd87391286 (hex) or
       0txM8kF0AjlAiUW9hzkShg== (base64).";
  }

  identity two-plus-n {
    if-feature "basic-link-maps";
    base link-map-patterns;
    description
      "Provides basic load-balancing on two links and any other
       links in the aggregation are standby. All packets with an
       even CID value are mapped to the link with the lowest link
       number that is active in the aggregation, and all packets
       with an odd CID value are mapped to the link with the
       highest link number that is active in the aggregation.
       This is updated whenever a link becomes active or inactive
       in the aggregation. The pattern supports any number of
       links with link numbers between 1 and 4095. Any links in
       the aggregation with a link number greater than 4095 are
       not used.
       The map consists of an identical entry for each even CID
       containing a list of numbers from 1 to 4095, followed by
       the number 0, and an identical entry for each odd CID
       containing a list of numbers from 4095 to 1, followed by
       the number 0.
       The MD5 digest calculated for this map is
       0xfe8fec2effe6cd1cbd78c2a968aa5d20 (hex) or
       /o/sLv/mzRy9eMKpaKpdIA== (base64).";
  }

  identity service-map-patterns {
    description
      "Base identity for patterns filling ~~the admin-conv-service-map.~~the
          admin-conv-service-map.  This identity is intended to ~~serve as base identity,
~~not~~serve
          as base identity, not to be directly referenced. Use of ~~the identity allows
vendors~~the
              identity allows vendors to augment the module with ~~vendor specified
patterns,";~~vendor
       specified patterns.";
    reference
      "7.3.1.1.36, 6.6.3.1 of IEEE Std 802.1AX";
  }

  typedef lacp-state {
    type bits {
      bit lacp-activity {
        position 1;
        description
          "Provides administrative control over when LACPDUs are
           transmitted.  A value of '1' indicates Active mode where
           LACPDUs are sent regardless of partner's lacp-activity
           value.  A value of '0' indicates Passive mode where
           LACPDUs are sent only when the partner's lacp-activity
           value is '1' (partner is in Active mode).";
```

```
        }
        bit lacp-timeout {
          position 2;
          description
            "Provides administrative control over the frequency of
             received LACPDUs.  A value of '1' indicates Short Timeout
             (so partner uses frequent transmission).  A value of '0'
             indicates Long Timeout (so partner can use infrequent
             transmission).";
        }
        bit aggregation {
          position 3;
          description
            "Provides administrative control over whether this
             Aggregation Port can be in a LAG with more than one
             member.  A value of '1' indicates the port can be
             aggregated with other ports.  A value of '0' indicates
             the port can only be a solitary link.";
        }
        bit synchronization {
          position 4;
          description
            "The Synchronization state of the MUX state machine.";
        }
        bit collecting {
          position 5;
          description
            "The Collecting state of the MUX state machine.";
        }
        bit distributing {
          position 6;
          description
            "The Distributing state of the MUX state machine.";
        }
        bit defaulted {
          position 7;
          description
            "Indicates the port is using the partner-admin values
             to select an Aggregator.";
        }
        bit expired {
          position 8;
          description
            "The Expired state of the Receive state machine.";
        }
      }
      description
        "LACP state values as transmitted in LACPDUs.
         Administrative control over the values of lacp-activity,
         lacp-timeout, aggregation, and (in partner-admin-lacp-state)
         synchronization is allowed.
         The remaining bits are read-only.";
      reference
        "6.4.1, 6.4.2.3 of IEEE Std 802.1AX";
    }

    typedef bridge-reserved-addresses {
      type enumeration {
        enum nearest-customer-bridge {
          description
            "Reserved MAC address 01-80-C2-80-00-00-00";
        }
        enum slow-protocols-multicast {
          description
            "Reserved MAC address 01-80-C2-80-00-00-02";
        }
        enum nearest-non-tpmr-bridge {
          description
            "Reserved MAC address 01-80-C2-80-00-00-03";
        }
        enum nearest-bridge {
          description
```

```
          "Reserved MAC address 01-80-C2-80-00-00-0E";
      }
    }
    description
      "Group addresses that are filtered by some or all types of
       bridges Bridges in order to limit the scope of propagation of PDUs
       in a Bridged Network.";
    reference
      "8.6.3 of IEEE Std 802.1Q";
  }

  grouping link-map {
    description
      "Specifies the contents of the admin-conv-link-map.";
    reference
      "7.3.1.1.34, 6.6.3.1 of IEEE Std 802.1AX";
    choice method {
      default "cid-list";
      description
        "Provides two ways to specify the map contents.";
      leaf pattern {
        type identityref {
          base link-map-patterns;
        }
        description
          "Use a predefined pattern to fill the map.";
      }
      list cid-list {
        key "cid";
        description
          "Data structure to map a Conversation Identifier
           (CID) to a Link Number. Each entry consists of a CID
           and a list of link numbers that can potentially be
           selected for that CID. An empty list of link-numbers
           means that no links are selected for the CID.";
        leaf cid {
          type uint16 {
            range "0..4095";
          }
          description
            "Port Conversation Identifier";
        }
        leaf-list link-numbers {
          type uint16;
          description
            "List of zero or more Link Numbers that can
             potentially be selected for distribution of frames
             with this CID.";
        }
      }
    }
  }

  grouping service-map {
    description
      "Specifies the contents of the admin-conv-service-map.";
    reference
      "7.3.1.1.36, 6.6.3.1 of IEEE Std 802.1AX";
    choice method {
      default "cid-list";
      description
        "Provides two ways to specify the map contents.";
      leaf pattern {
        type identityref {
          base service-map-patterns;
        }
        description
          "Use a predefined pattern to fill the map.";
      }
      list cid-list {
        key "cid";
        description
```

```
          "Data structure to map service identifiers Service Identifiers to
           conversation identifiersConversation Identifiers.  Each entry consists of a
           Conversation ID (CID) and a list of zero or more
           Service Identifiers (SIDs) that map to it. An empty
           list of SIDs means there are no SIDs that map to this
           CID, and results in the same behavior as not having an
           entry for this CID.";
        leaf cid {
          type uint16 {
            range "0..4095";
          }
          description
            "Port Conversation Identifier";
        }
        leaf-list service-ids {
          type uint32;
          description
            "List of zero or more SIDs that map to the CID.";
        }
      }
    }
  }
}
```

10.7.2 The *ieee802-dot1ax-linkagg* YANG module

```
module ieee802-dot1ax-linkagg {
  yang-version 1.1;
  namespace "urn:ieee:std:802.1AX:yang:ieee802-dot1ax-linkagg";
  prefix dot1ax;

  import ieee802-dot1ax-types {
    prefix ax;
  }
  import ieee802-types {
    prefix ieee;
  }
  import ietf-yang-types {
    prefix yang;
  }
  import ietf-interfaces {
    prefix if;
  }
  import iana-if-type {
    prefix ianaif;
  }

  organization
    "IEEE 802.1 Working Group";
  contact
    "WG-URL: http://www.ieee802.org/1/
     WG-Email: stds-802-1-l@ieee.org

     Contact: IEEE 802.1 Working Group Chair
      Postal: C/O IEEE 802.1 Working Group
              IEEE Standards Association
              45 Hoes Lane
              Piscataway, NJ  08854
              USA
      E-mail: stds-802-1-chairs@ieee.org";
  description
    "This YANG module describes the configuration model for Link
     Aggregation, as specified in IEEE Std 802.1AX, including Link
     Aggregation Control Protocol (LACP) and Conversation Conversation-Sensitive
     Collection and Distribution.

     Copyright (C) IEEE (2025).

     This version of this YANG module is part of IEEE Std 802.1AX;
```

```
      see the standard itself for full legal notices.";

  revision 2025-09-~~05~~30 {
    description
      "Published as part of IEEE Std 802.1AXdz-2025.
       The following reference statement identifies each ~~referenced IEEE~~referenced
       IEEE Standard as updated by applicable amendments.";
    reference
      "IEEE Std 802.1AX-2020 Link Aggregation:
       IEEE Std 802.1AXdz-2025.";
  }

  typedef key-group-ref {
    type leafref {
      path "/dot1ax:linkagg/dot1ax:key-group/dot1ax:name";
    }
    description
      "This type is used by ~~aggregators~~ Aggregators and ~~aggregation ports~~ Aggregation Ports
to
       reference an entry in the key-group list.";
  }

  typedef agg-system-ref {
    type leafref {
      path "/dot1ax:linkagg/dot1ax:agg-system/dot1ax:name";
    }
    description
      "This type is used in the key-group list entries to
       reference an entry in the agg-system list.";
  }

  augment "/if:interfaces/if:interface" {
    when "derived-from-or-self(if:type,'ianaif:ieee8023adLag')" {
      description
        "Applies to interfaces representing a LAG.";
    }
    description
      "Augment Interface with Aggregator parameters.";
    container lag {
      description
        "Contains the configuration and status information that
         allows an instance of an Aggregator to be managed.";
      leaf key-group-name {
        type key-group-ref;
        mandatory true;
        description
          "Specifies the entry in the link-aggregation key-group
           list to which this ~~aggregator~~ Aggregator is assigned.";
      }
      leaf mac-address {
        type ieee:mac-address;
        config false;
        description
          "The MAC address assigned to the Aggregator.";
        reference
          "7.3.1.1.9 of IEEE Std 802.1AX";
      }
      container lacp {
        config false;
        description
          "Contains ~~aggregator~~ Aggregator LACP operational data.";
        leaf actor-oper-key {
          type uint16;
          description
            "The current operational value of the Key for the
             Aggregator. The meaning of particular Key
             values is of local significance.";
          reference
            "7.3.1.1.8 of IEEE Std 802.1AX";
        }
        leaf partner-system {
          type ieee:mac-address;
```

```
      description
        "The value of the MAC address portion of the
         System Identifier for the current
         protocol partner of this Aggregator. A value
         of zero indicates that there is no known partner.
         If the aggregation is manually configured, this
         System ~~identifier~~ Identifier value is assigned by the
         local System.";
      reference
        "7.3.1.1.10 of IEEE Std 802.1AX";
    }
    leaf partner-system-priority {
      type uint16;
      description
        "Indicates the priority value portion of the current
         Partner's System Identifier. If the aggregation is
         manually configured, this System Priority value is
         assigned by the local System.";
      reference
        "7.3.1.1.11 of IEEE Std 802.1AX";
    }
    leaf partner-oper-key {
      type uint16;
      description
        "The current operational value of the Key for the
         ~~Aggregators~~ Aggregator's current protocol Partner. If the
         aggregation is manually configured, this Key value
         is assigned by the local System.";
      reference
        "7.3.1.1.12 of IEEE Std 802.1AX";
    }
    leaf aggregate-or-individual {
      type boolean;
      description
        "Indicates whether the Aggregator represents an
         Aggregate (TRUE) or an Individual link (FALSE).";
      reference
        "7.3.1.1.6 of IEEE Std 802.1AX";
    }
  }
  container cscd {
    if-feature "ax:cscd";
    config false;
    description
      "Aggregator parameters obtained by the operation of LACP
       supporting CSCD.";
    leaf partner-port-algorithm {
      type identityref {
        base ax:distribution-algorithm;
      }
      description
        "Operational value of the distribution algorithm in
         use by the LACP Partner.";
      reference
        "7.3.1.1.40 of IEEE Std 802.1AX";
    }
    leaf partner-conv-service-digest {
      type binary;
      description
        "The MD5 Digest of the admin-conv-service-map in use
         by the LACP Partner.";
      reference
        "7.3.1.1.42, 6.6.3.1 of IEEE Std 802.1AX";
    }
    leaf partner-conv-link-digest {
      type binary;
      description
        "The MD5 Digest of the admin-conv-link-map in use
         by the LACP Partner.";
      reference
        "7.3.1.1.41, 6.6.3.1 of IEEE Std 802.1AX";
    }
```

```
        leaf discard-wrong-conversation {
          type boolean;
          description
            "The operational value that determines whether an
             Aggregator discards a frame that is collected
             from an Aggregation Port that is different from the
             Aggregation Port to which the Aggregator would
             distribute a frame with the same Port Conversation
             ID.";
          reference
            "7.3.1.1.37, 6.6 of IEEE Std 802.1AX";
        }
        leaf-list active-lag-links {
          type uint16;
          description
            "A list, possibly empty, of the operational
             link-number of each Aggregation Port active
             (i.e~~. ~~., Collecting) on this Aggregator.";
          reference
            "7.3.1.1.43 of IEEE Std 802.1AX";
        }
      }
    }
  }

  augment "/if:interfaces/if:interface/if:statistics" {
    when '../dot1ax:lag' {
      description
        "Applies to ~~aggregators~~Aggregators.";
    }
    description
      "Augment interface statistics with ~~aggregator ~~Aggregator statistics.";
    container lag-stats {
      config false;
      description
        "Contains the set of stats associated with the
         Aggregator.";
      leaf unknown-protocol-frames {
        type yang:counter64;
        description
          "A count of data frames discarded on reception by all
           ports that are (or have been) members of the
           aggregation, due to the detection of an unknown Slow
           Protocols PDU (7.3.3.1.5)";
        reference
          "7.3.1.1.29 of IEEE Std 802.1AX";
      }
    }
  }

  augment "/if:interfaces/if:interface" {
    description
      "Augment interface model with Aggregation ~~port~~Port
       configuration nodes.";
    container aggport {
      presence
        "When present, this interface supports Link Aggregation";
      description
        "Contains Aggregation Port configuration related nodes,
         which provides the basic management controls necessary
         to allow an instance of an Aggregation Port to be managed,
         for the purposes of Link Aggregation.";
      leaf key-group-name {
        type key-group-ref;
        mandatory true;
        description
          "Specifies the entry in the link-aggregation key-group
           list to which this aggregation-port is assigned.";
      }
      leaf actor-port-number {
        type uint16 {
          range "1..65535";
```

```
    }
    description
      "The port number assigned to the Aggregation Port.
       The port number is communicated in LACPDUs as the
       Actor_Port.
       This leaf provides the ability to administratively
       override the initial value provided by the system.";
    reference
      "7.3.2.1.14, 6.4.6 of IEEE Std 802.1AX";
  }
  leaf actor-port-priority {
    type uint16;
    default "0x8000";
    description
      "The priority value assigned to this Aggregation Port.";
    reference
      "7.3.2.1.15, 6.4.6 of IEEE Std 802.1AX";
  }
  leaf partner-admin-key {
    type uint16 {
      range "1..65535";
    }
    description
      "The current administrative value of the Key for the
       protocol Partner. The assigned value is used, along
       with the value of port-partner-admin-system-priority,
       partner-admin-system, partner-admin-port, and
       partner-admin-port-priority, in order to achieve
       manually configured aggregation.
       The default value is the initial value of actor-port-
       number provided by the system.";
    reference
      "7.3.2.1.10 of IEEE Std 802.1AX";
  }
  leaf partner-admin-port {
    type uint16 {
      range "1..65535";
    }
    description
      "The current administrative value of the port number for
       the protocol Partner. The assigned value is used, along
       with the value of partner-admin-system-priority,
       partner-admin-system, port-partner-admin-key, and
       partner-admin-port-priority, in order to achieve
       manually configured aggregation.
       The default value is the initial value of actor-port-
       number provided by the system.";
    reference
      "7.3.2.1.16 of IEEE Std 802.1AX";
  }
  leaf partner-admin-port-priority {
    type uint16;
    default "0";
    description
      "The current administrative value of the port priority
       for the protocol Partner. The assigned value is used,
       along with the value of partner-admin-system-priority,
       partner-admin-system, partner-admin-key, and
       partner-admin-port, in order to achieve manually
       configured aggregation.";
    reference
      "7.3.2.1.18 of IEEE Std 802.1AX";
  }
  leaf partner-admin-state {
    type ax:lacp-state {
      bit lacp-activity {
        position 1;
        description
          "When the LACP_Activity state is set to '0', the
           transmission of LACPDUs is controlled by the
           actor-admin-state.LACP_Activity.";
      }
```

```
        bit lacp-timeout {
          position 2;
          description
            "When the LACP_Timeout is set to '0', LACPDUs
             are transmitted at the Slow_Periodic_Time.";
        }
        bit aggregation {
          position 3;
          description
            "When the Aggregation state is set to '0', this
             port cannot be aggregated with any other port.";
        }
        bit synchronization {
          position 4;
          description
            "When the Synchronization state is set to '0', this
             port cannot become active.";
        }
      }
      default "synchronization";
      description
        "Provides administrative control over the partner's
         LACP_Activity, LACP_Timeout, Aggregation, and
         Synchronization state when the partner's ~~information is~~information
         is unknown (i.e., no LACPDUs are received from the
         ~~unknown (i.e. no LACPDUs are received from the ~~partner).";
      reference
        "7.3.2.1.22, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
    }
    leaf wtr-time {
      type uint16 {
        range "0..32767";
      }
      default "0";
      description
        "The wait-to-restore (WTR) period, in seconds, that
         needs to elapse between an Aggregation Port on a LAG
         coming up (Port_Operational becoming TRUE) and being
         permitted to become active (transmitting and
         receiving frames) on the LAG.";
      reference
        "7.3.2.1.30 of IEEE Std 802.1AX";
    }
    leaf wtr-revertive {
      type boolean;
      default "true";
      description
        "Controls revertive or non-revertive mode of operation.
         When TRUE, the Aggregation Port can become active as
         soon as the wait-to-restore timer expires regardless of
         the state of other links in the LAG.
         When FALSE, the Aggregation Port cannot become active
         unless there are no other links that can become active
         in the LAG. The default value is TRUE.";
      reference
        "7.3.2.1.31 of IEEE Std 802.1AX";
    }
    container lacp {
      config false;
      description
        "Contains Aggregation ~~port ~~Port LACP operational related
         nodes.";
      leaf actor-lacp-version {
        type uint8;
        description
          "The version number transmitted in LACPDUs on this
           Aggregation Port";
        reference
          "7.3.2.1.33 of IEEE Std 802.1AX";
      }
      leaf actor-oper-key {
        type uint16;
```

```
  description
    "The current operational value of the Key for the
     Aggregation Port. The meaning of particular Key values
     is of local significance.";
  reference
    "7.3.2.1.5 of IEEE Std 802.1AX";
}
leaf actor-oper-state {
  type ax:lacp-state;
  description
    "The operational value of the Actor_State as
     transmitted in LACPDUs.";
  reference
    "7.3.2.1.21, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
}
leaf partner-lacp-version {
  type uint8;
  description
    "The version number in the LACPDU most recently
     received on this Aggregation Port.";
  reference
    "7.3.2.1.34 of IEEE Std 802.1AX";
}
leaf partner-oper-system-priority {
  type uint16;
  description
    "Indicates the operational value of priority associated
     with the Partners System ID. The value of this
     attribute can contain the manually configured value
     carried in partner-admin-system-priority if there is
     no protocol Partner.";
  reference
    "7.3.2.1.7 of IEEE Std 802.1AX";
}
leaf partner-oper-system {
  type ieee:mac-address;
  description
    "Represents the MAC address portion of the Aggregation
     Port's current protocol partner's System Identifier.
     A value of zero indicates that there is no known
     protocol Partner.
     The value of this attribute can contain the manually
     configured value carried in partner-admin-system if
     there is no protocol Partner.";
  reference
    "7.3.2.1.9 of IEEE Std 802.1AX";
}
leaf partner-oper-key {
  type uint16;
  description
    "The current operational value of the Key for the
     protocol Partner. The value of this attribute can
     contain the manually configured value carried in
     partner-admin-key if there is no protocol Partner.";
  reference
    "7.3.2.1.11 of IEEE Std 802.1AX";
}
leaf partner-oper-port {
  type uint16;
  description
    "The operational port number assigned by the
     Aggregation Port's protocol Partner. The value of this
     attribute can contain the administratively configured
     value carried in partner-admin-port if there is no
     protocol Partner.";
  reference
    "7.3.2.1.17 of IEEE Std 802.1AX";
}
leaf partner-oper-port-priority {
  type uint16;
  description
    "The operational priority value assigned by the
```

```
          Aggregation Port's protocol Partner. The value of this
          attribute can contain the administratively configured
          value carried in partner-admin-port-priority if there
          is no protocol Partner.";
      reference
        "7.3.2.1.19 of IEEE Std 802.1AX";
    }
    leaf partner-oper-state {
      type ax:lacp-state;
      description
        "The operational value of the partner's LACP state
         derived from received LACPDUs or, when Defaulted is
         true, from the partner-admin-state.";
      reference
        "7.3.2.1.23, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
    }
    leaf aggregate-or-individual {
      type boolean;
      description
        "When true indicates the Aggregation Port can join a
         LAG consisting of multiple Aggregation Ports.
         When false, indicates that the Aggregation Port can
         only operate as an Solitary link because the
         Aggregation bit is false in either
         actor-oper-port-state or partner-oper-port-state.";
      reference
        "7.3.2.1.24 of IEEE Std 802.1AX";
    }
    leaf wtr-waiting {
      type boolean;
      description
        "Indicates the Aggregation Port is inhibited from
         becoming active for an interval (determined by
         wtr-time) after becoming operational or while
         non-revertive operation is being enforced by the
         Selection Logic.";
      reference
        "7.3.2.1.32 of IEEE Std 802.1AX";
    }
  }
  container cscd {
    if-feature "ax:cscd";
    description
      "Aggregation Port parameters for support of CSCD.";
    leaf admin-link-number {
      type uint16;
      description
        "The Link_Number value for the Aggregation Port,
         that is unique among all Aggregation Ports in the same
         key group. This leaf provides the ability to
         administratively override the initial value provided
         by the system. A value of 0 is allowed, but
         results in no frames being distributed to this
         Aggregation Port.
         The default value is the initial value of actor-port-
         number provided by the system.";
      reference
        "7.3.2.1.27, 6.6.3.2 of IEEE Std 802.1AX";
    }
    leaf partner-link-number {
      type uint16;
      config false;
      description
        "The last received value of the Partner_Link_Number,
         or zero if the Aggregation Port is using default
         values for the Partner or the Partner LACP Version
         is 1.";
      reference
        "7.3.2.1.29 of IEEE Std 802.1AX";
    }
    leaf link-number {
      type uint16;
```

```
          config false;
          description
            "The operational link number for this Aggregation Port.
             The value is either the same as the admin-link-number,
             or the corresponding value of the LACP partner.";
          reference
            "7.3.2.1.28 of IEEE Std 802.1AX";
        }
      }
    }
  }

  augment "/if:interfaces/if:interface/if:statistics" {
    when '../dot1ax:aggport' {
      description
        "Applies to ~~aggregation ports~~Aggregation Ports.";
    }
    description
      "Augment interface statistics with aggport statistics.";
    container aggport-stats {
      config false;
      description
        "Contains stats associated with the Aggregation Port.";
      leaf lacp-pdu-rx {
        type yang:counter64;
        description
          "The number of valid LACPDUs received on this
           Aggregation Port.";
        reference
          "7.3.3.1.2 of IEEE Std 802.1AX";
      }
      leaf marker-pdu-rx {
        type yang:counter64;
        description
          "The number of valid Marker PDUs received on this
           Aggregation Port.";
        reference
          "7.3.3.1.3 of IEEE Std 802.1AX";
      }
      leaf marker-response-pdu-rx {
        type yang:counter64;
        description
          "The number of valid Marker Response PDUs received on
           this Aggregation Port.";
        reference
          "7.3.3.1.4 of IEEE Std 802.1AX";
      }
      leaf unknown-rx {
        type yang:counter64;
        description
          "The number of frames received that either:
           a) Carry the Slow Protocols Ethernet Type value,
           but contain an unknown PDU, or
           b) Are addressed to the Slow Protocols group MAC
           Address, but do not carry the Slow Protocols
           Ethernet Type.";
        reference
          "7.3.3.1.5 of IEEE Std 802.1AX";
      }
      leaf illegal-rx {
        type yang:counter64;
        description
          "The number of frames received that carry the Slow
           Protocols Ethernet Type value, but contain a badly
           formed PDU or an illegal value of Protocol Subtype.";
        reference
          "7.3.3.1.6 of IEEE Std 802.1AX";
      }
      leaf lacp-pdu-tx {
        type yang:counter64;
        description
          "The number of LACPDUs transmitted on this
```

```
             Aggregation Port.";
          reference
            "7.3.3.1.7 of IEEE Std 802.1AX";
        }
        leaf marker-pdu-tx {
          type yang:counter64;
          description
            "The number of Marker PDUs transmitted on this
             Aggregation Port.";
          reference
            "7.3.3.1.8 of IEEE Std 802.1AX";
        }
        leaf marker-response-pdu-tx {
          type yang:counter64;
          description
            "The number of Marker Response PDUs transmitted on
             this Aggregation Port.";
          reference
            "7.3.3.1.9 of IEEE Std 802.1AX";
        }
      }
    }

    container linkagg {
      description
        "Link Aggregation System specific configuration nodes.";
      list agg-system {
        key "name";
        description
          "List of ~~aggregation systems~~Aggregation Systems.";
        leaf name {
          type string;
          description
            "Name for the ~~aggregation system~~Aggregation System.";
        }
        leaf actor-system {
          type ieee:mac-address;
          description
            "The part of the System Identifier that is a globally
             unique MAC address.  This leaf provides the ability to
             administratively override the initial value provided
             by the system.";
          reference
            "7.3.1.1.4, 7.3.2.1.3 of IEEE Std 802.1AX";
        }
        leaf actor-system-priority {
          type uint16;
          default "0x8000";
          description
            "The part of the System Identifier that is the
             priority of the system.";
          reference
            "7.3.1.1.5, 7.3.2.1.2  of IEEE Std 802.1AX";
        }
      }
      list key-group {
        key "name";
        unique "actor-admin-key";
        description
          "List of key groups.  A key group is the set of ~~aggregators~~Aggregators
           and ~~aggregation ports~~ Aggregation Ports that share the same system priority,
           system identifier, and aggregation key, and therefore can
           potentially form a Link Aggregation Group.  Each entry in
           the key group list contains the parameters common to all
             ~~aggregation ports~~ Aggregation Ports and/or ~~aggregators~~ Aggregators in the key
group.";
        reference
          "6.3.5, 6.4.12 of IEEE Std 802.1AX";
        leaf name {
          type string;
          description
            "Name for the key group.";
```

```
}
leaf actor-admin-key {
  type uint16 {
    range "1..65535";
  }
  mandatory true;
  description
    "The administrative value of the key used by the
     Aggregators and Aggregation Ports in this key-group.";
  reference
    "7.3.1.1.7, 7.3.2.1.4 of IEEE Std 802.1AX";
}
leaf agg-system-name {
  type agg-system-ref;
  mandatory true;
  description
    "Specifies the ~~aggregation system ~~Aggregation System for this key
     group.";
}
leaf actor-protocol-da {
  type ax:bridge-reserved-addresses;
  must ". = 'nearest-customer-bridge' or
        . = 'slow-protocols-multicast' or
        . = 'nearest-non-tpmr-bridge'" {
    error-message "Invalid protocol address";
  }
  default "slow-protocols-multicast";
  description
    "A 6-octet read-write MAC Address value specifying the DA
     to be used when sending Link Aggregation Control and
     Marker PDUs.  Valid addresses are the
     nearest-customer-bridge, slow-protocols-multicast,
     and nearest-non-tpmr-bridge group addresses.  The
     default value is the slow-protocols-multicast address.";
  reference
    "7.3.2.2.1, 6.2.10.2 of IEEE Std 802.1AX";
}
leaf collector-max-delay {
  type uint16;
  description
    "Specifies the maximum delay, in tens of microseconds,
     between receiving a frame from an Aggregator Port, and
     either delivering the frame to the Aggregator Client or
     discarding the frame. A value of zero means the delay
     is less than the minimum increment (< 10us).
     This leaf provides the ability to administratively
     override the initial value provided by the system.";
  reference
    "7.3.1.1.32, 6.2.3.1.1, B.3 of IEEE Std 802.1AX";
}
leaf actor-admin-state {
  type ax:lacp-state {
    bit lacp-activity {
      position 1;
      description
        "Setting the LACP_Activity state to '0' means that the
         transmission of LACPDUs is controlled by the value of
         partner-oper-state.LACP_Activity.";
    }
    bit lacp-timeout {
      position 2;
      description
        "Setting the LACP_Timeout to '0' means that actor uses
         the Long_Timeout value, allowing the partner ~~to transmit~~to
         transmit LACPDUs at the Slow_Periodic_Time.";
    }
    bit aggregation {
      position 3;
      description
        "Setting the Aggregation state to '0' prevents this
         port from being aggregated with any other ports.";
    }
```

```
    }
    default "lacp-activity lacp-timeout aggregation";
    description
      "Provides administrative control over the values of the
       LACP_Activity, LACP_Timeout, and Aggregation state.";
    reference
      "7.3.2.1.20, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
  }
  leaf partner-admin-system {
    type ieee:mac-address;
    default "00-00-00-00-00-00";
    description
      "The administrative value of the MAC address portion of
       the Partner's System Identifier.
       The assigned value is used, along with the value of
       port-partner-admin-system, partner-admin-key,
       partner-admin-port, and partner-admin-port-priority,
       to achieve administratively configured Link
       Aggregation Groups with a partner that does not run
       LACP.";
    reference
      "7.3.2.1.8 of IEEE Std 802.1AX";
  }
  leaf partner-admin-system-priority {
    type uint16;
    default "0";
    description
      "The administrative value of priority portion of the
       the Partner's System Identifier. The assigned
       value is used, along with the value of
       port-partner-admin-system, partner-admin-key,
       partner-admin-port, and partner-admin-port-priority,
       to achieve administratively configured Link
       Aggregation Groups with a partner that does not run
       LACP.";
    reference
      "7.3.2.1.6 of IEEE Std 802.1AX";
  }
  leaf port-algorithm {
    type identityref {
      base ax:distribution-algorithm;
    }
    default "ax:unspecified";
    description
      "Identifies the algorithm used by the Aggregator to
       assign frames to a Port Conversation ID.  Default is
       the value for an unspecified distribution algorithm.
       When the identity description specifies a ~~4 ~~4-octet value,
       this value will be used for ~~csed ~~CSCD purposes, and included
       in LACPDUs. Otherwise this leaf is used purely for
       local selection of a distribution algorithm, and the
       unspecified distribution algorithm is used for ~~csed~~CSCD
       purposes.";
    reference
      "7.3.1.1.33, Table 8-1 of IEEE Std 802.1AX";
  }
  leaf-list lags {
    type if:interface-ref;
    config false;
    description
      "A list of the if:name of ~~aggregators ~~Aggregators assigned to this
       key group.";
    reference
      "linkagg:key-group";
  }
  leaf-list aggports {
    type if:interface-ref;
    config false;
    description
      "A list of the if:name of ~~aggregation ports ~~Aggregation Ports assigned to
       this key group.";
    reference
```

```
          "linkagg:key-group";
      }
      container cscd {
        if-feature "ax:cscd";
        description
          "Contains CSCD parameters that need to be consistent for
           all ~~aggregation ports ~~Aggregation Ports and ~~aggregators ~~Aggregators in the key
group.";
        container admin-conv-service-map {
          if-feature "ax:sid-map";
          description
            "Data structure to map ~~service identifiers ~~Service Identifiers to
             ~~conversation identifiers~~Conversation Identifiers.  Each entry consists of a
             Conversation ID (CID) and a list of zero or more
             Service Identifiers (SIDs) that map to it. An empty
             list of SIDs means there are no SIDs that map to this
             CID, and results in the same behavior as not having an
             entry for this CID.";
          reference
            "7.3.1.1.36, 6.6.3.1 of IEEE Std 802.1AX";
          uses ax:service-map;
        }
        leaf admin-conv-service-digest {
          type binary;
          config false;
          description
            "The MD5 Digest of the admin-conv-service-map. The
             value is NULL (AAAAAAAAAAAAAAAAAAAAAA== in base64)
             when the distribution algorithm specified by
             agg-port-algorithm does not use the
             admin-conv-service-map.";
          reference
            "7.3.1.1.39, 6.6.3.1 of IEEE Std 802.1AX";
        }
        container admin-conv-link-map {
          description
            "Data structure to map a Conversation Identifier
             (CID) to a Link Number. Each entry consists of a CID
             and a list of link numbers that can potentially be
             selected for that CID. An empty list of link-numbers
             means that no links are selected for the CID.";
          reference
            "7.3.1.1.34, 6.6.3.1 of IEEE Std 802.1AX";
          uses ax:link-map;
        }
        leaf admin-conv-link-digest {
          type binary;
          config false;
          description
            "The MD5 Digest of the admin-conv-link-map. The
             value is NULL (AAAAAAAAAAAAAAAAAAAAAA== in base64)
             when the distribution algorithm specified by
             agg-port-algorithm does not use the
             admin-conv-link-map.";
          reference
            "7.3.1.1.38, 6.6.3.1 of IEEE Std 802.1AX";
        }
        leaf admin-discard-wrong-conv {
          if-feature "ax:dwc";
          type enumeration {
            enum force-true {
              value 1;
              description
                "Indicates that an Aggregator discards a
                 frame that is collected from an Aggregation Port
                 that is different from the Aggregation Port to
                 which the Aggregator would distribute a frame
                 with the same Port Conversation ID.";
            }
            enum force-false {
              value 2;
              description
```

```
                "Indicates that an Aggregator does not discard
                 a frame that is collected from an Aggregation Port
                 that is different from the Aggregation Port to
                 which the Aggregator would distribute a frame with
                 the same Port Conversation ID. This is the behavior
                 of the Aggregator when DWC is not supported";
            }
            enum auto {
              value 3;
              description
                "Indicates that the Aggregator behaves as
                 if the value was force-true only when the actor
                 and partner agree on the algorithms (other than
                 unspecified) and mapping tables used to map frames
                 to Aggregation Ports, and behaves as if the value
                 was force-false otherwise.";
            }
          }
          default "force-false";
          description
            "Indicates whether an Aggregator discards a
             frame that is collected from an Aggregation Port
             that is different from the Aggregation Port to which
             the Aggregator would distribute a frame with the
             same Port Conversation ID.";
          reference
            "7.3.1.1.35, 6.6 of IEEE Std 802.1AX";
        }
      }
    }
  }
}
```

10.7.3 The *ieee802-dot1ax-drni* YANG module

```
module ieee802-dot1ax-drni {
  yang-version 1.1;
  namespace "urn:ieee:std:802.1AX:yang:ieee802-dot1ax-drni";
  prefix dot1ax-drni;

  import ieee802-dot1ax-types {
    prefix ax;
  }
  import ieee802-dot1ax-linkagg {
    prefix dot1ax;
  }
  import ieee802-types {
    prefix ieee;
  }
  import ietf-yang-types {
    prefix yang;
  }
  import ietf-interfaces {
    prefix if;
  }

  organization
    "IEEE 802.1 Working Group";
  contact
    "WG-URL: http://www.ieee802.org/1/
     WG-Email: stds-802-1-l@ieee.org

     Contact: IEEE 802.1 Working Group Chair
      Postal: C/O IEEE 802.1 Working Group
              IEEE Standards Association
              45 Hoes Lane
              Piscataway, NJ  08854
              USA
      E-mail: stds-802-1-chairs@ieee.org";
```

```
  description
    "This YANG module describes the configuration model for a
     Distributed Resilient Network Interface (DRNI) as specified
     in 802.1AX.

     Copyright (C) IEEE (2025).

     This version of this YANG module is part of IEEE Std 802.1AX;
     see the standard itself for full legal notices.";

  revision 2025-09-~~05 ~~30 {
    description
      "Published as part of IEEE Std 802.1AXdz-2025.
       The following reference statement identifies each ~~referenced IEEE~~referenced
       IEEE Standard as updated by applicable amendments.";
    reference
      "IEEE Std 802.1AX-2020 Link Aggregation:
       IEEE Std 802.1AXdz-2025.";
  }

  typedef irp-state {
    type bits {
      bit reserved-1 {
        position 1;
        description
          "Reserved for future use. It is set to 0 on
           transmit and ignored on receipt.";
      }
      bit reserved-2 {
        position 2;
        description
          "Reserved for future use. It is set to 0 on
           transmit and ignored on receipt.";
      }
      bit short-timeout {
        position 3;
        description
          "The Short_Timeout flag indicates the Timeout ~~control value~~control
           value in use by the DRCP Receive machine on this IRP~~. Short Timeout~~.
           Short Timeout is encoded as a 1; Long Timeout is encoded
           ~~is encoded as a 1; Long Timeout is encoded ~~as a 0.";
      }
      bit synchronization {
        position 4;
        description
          "When the Sync flag is TRUE (1), the DRCP Receive ~~machine has~~machine
           has determined the Neighbor DRNI System has a compatible
           configuration for forming a DRNI.";
      }
      bit irc-data {
        position 5;
        description
          "When the IRC_Data flag is TRUE (1), the transfer of Up
           and Down frames is permitted on the IRC.";
      }
      bit drni {
        position 6;
        description
          "The DRNI flag is TRUE (1) when this DRNI System is paired
           with another DRNI System and FALSE (0) otherwise.";
      }
      bit defaulted {
        position 7;
        description
          "When the Defaulted flag is TRUE (1), the DRCP ~~Receive machine~~Receive
           machine is using default operational ~~Neighbor information.~~Neighbor
           information. When FALSE (0), the operational ~~Neighbor information~~Neighbor
           information in use has been received in a DRCPDU.";
      }
      bit expired {
        position 8;
        description
```

```
          "When the Expired flag is TRUE (1), the DRCP ~~Receive machine~~Receive
           machine is in the EXPIRED state.";
      }
    }
    description
      "IRP state values as transmitted in DRCPDUs.
       Administrative control over the values of Short_Timeout
       and IRC_Data is allowed. The remaining bits are read-only.";
    reference
      "7.4.1.1.24, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
  }

  grouping drni-mask {
    description
      "Specifies the contents of a bit mask indexed by CID.";
    leaf-list cid-str {
      type string {
        pattern '(0|[1-9][0-9]{0,3})'
              + '((-(0|[1-9][0-9]{0,3})'
              + '(%(0|[1-9][0-9]{0,3}))?)?)'
              + '(,'
              + '(0|[1-9][0-9]{0,3})'
              + '((-(0|[1-9][0-9]{0,3})'
              + '(%(0|[1-9][0-9]{0,3}))?)?)'
              + ')*';
      }
      default "0-4095";
      description
        "A comma separated list of single CID values (x),
         CID ranges (x-y), or ranges with an increment (x-y%z).
         For example, the string '0,10-13,100-110%4' sets the
         bits in the mask corresponding to CIDs
         0,10,11,12,13,100,104,108.
         Some special cases are:
          'x-y%z' where x+z>y is the same as 'x'.
          'x-y%0' is the same as 'x'.
          'x-y%1' is the same as 'x-y'.
          'x-y%2' sets the bit for all even CIDs in the range
            when x is even, and for all odd CIDs when x is odd.
         Although the pattern allows values of x, y, and z in the
         range 0-9999, resulting CID values greater than 4095
         have no effect on the mask.
         Specified as a list of strings, so that long strings
         can be broken into several shorter strings.
         The default sets all bits in the mask.";
    }
    leaf invert-mask {
      type boolean;
      default "false";
      description
        "When true the specified mask is inverted:
         each zero replaced with a one,
         and each one replaced with a zero.";
    }
  }

  augment "/if:interfaces/if:interface/dot1ax:lag" {
    description
      "Augmentation parameters only for Aggregators with
       DRNI enabled.";
    container drni {
      presence "When present, this Aggregator is enabled for DRNI";
      description
        "Aggregator parameters to support a Distributed
         Resilient Network Interface";
      leaf irp-name {
        type if:interface-ref;
        mandatory true;
        description
          "Interface Name (if:name) of the Port supporting the
           ~~Intra-~~Intra-Relay Port (IRP) of this DRNI Gateway.";
        reference
```

```
            "7.4.1.1.23 of IEEE Std 802.1AX";
        }
        leaf drni-aggregator-key {
          type uint16 {
            range "1..65535";
          }
          description
            "The Aggregator Key value to be used by the Aggregator
             supporting this DRNI Gateway (and the Aggregation Ports
             assigned to this DRNI Gateway) when paired with a
             neighbor DRNI System via the IRC. The default value is
             the Aggregator~~’~~'s Actor_Admin_Aggregator_Key value.";
          reference
            "7.4.1.1.15 of IEEE Std 802.1AX";
        }
        leaf drni-aggregator-system {
          type ieee:mac-address;
          default "00-00-00-00-00-00";
          description
            "The Aggregator System value to be used by the
             Aggregator supporting this DRNI Gateway (and the
             Aggregation Ports assigned to this DRNI Gateway)
             when paired with a neighbor DRNI System via the
             Intra-Relay Connection (IRC).";
          reference
            "7.4.1.1.13 of IEEE Std 802.1AX";
        }
        leaf drni-aggregator-system-priority {
          type uint16;
          default "0";
          description
            "The Aggregator System Priority value to be used by the
             Aggregator supporting this DRNI Gateway (and the
             Aggregation Ports assigned to this DRNI Gateway) when
             paired with a neighbor DRNI System via the IRC.";
          reference
            "7.4.1.1.14 of IEEE Std 802.1AX";
        }
        leaf drcp-protocol-da {
          type ax:bridge-reserved-addresses;
          must ". = 'nearest-customer-bridge' or
                . = 'nearest-bridge' or
                . = 'nearest-non-tpmr-bridge'" {
            error-message "Invalid protocol address";
          }
          default "nearest-non-tpmr-bridge";
          description
            "A 6-octet read-write MAC Address value specifying the
             Destination Address for Distributed Relay Control PDUs
             transmitted on the Intra-Relay Port. Valid addresses are
             the nearest-customer-bridge, nearest-bridge, and
             nearest-non-tpmr-bridge group addresses.  The default
             value is the nearest-non-tpmr-bridge group
             address.";
          reference
            "7.4.1.1.12, 9.6.1.1 of IEEE Std 802.1AX";
        }
        leaf home-admin-irp-state {
          type irp-state {
            bit short-timeout {
              position 3;
              description
                "The Short Timeout flag indicates the Timeout control
                 ~~"The Short_Timeout flag indicates~~ value in use by the ~~Timeout control value~~DRCP Receive machine on this
                 ~~in use by the DRCP Receive machine on this~~ IRP. Short Timeout is encoded as a 1; Long Timeout
                 is encoded as a ~~1; Long Timeout is encoded as a~~ 0.";
            }
            bit irc-data {
              position 5;
              description
```

```
        "When the IRC_Data flag is TRUE (1), the transfer ~~of Up~~of
         Up and Down frames is permitted on the IRC.";
    }
  }
  default "short-timeout irc-data";
  description
    "Provides administrative control over the values of the
     Short_Timeout and IRC-Data state.";
  reference
    "7.4.1.1.24, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
}
leaf home-oper-irp-state {
  type irp-state;
  config false;
  description
    "A string of 8 bits, corresponding to the current
     operational value of IRP_State as transmitted in
     DRCPDUs.";
  reference
    "7.4.1.1.25, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
}
leaf home-cscd-gateway-control {
  type boolean;
  default "true";
  description
    "When TRUE, allows the DRNI Gateway Port selection to
     be based on the CSCD parameters that control the
     Aggregator Port selection.";
  reference
    "7.4.1.1.16 of IEEE Std 802.1AX";
}
leaf home-dr-client-gateway-control {
  type boolean;
  default "true";
  description
    "When TRUE, allows the Distributed Relay Client to
     determine whether to forward frames through the DRNI
     Gateway Port.";
  reference
    "7.4.1.1.17 of IEEE Std 802.1AX";
}
leaf home-gateway-algorithm {
  type identityref {
    base ax:distribution-algorithm;
  }
  default "ax:unspecified";
  description
    "Identifies the algorithm used by the DRNI Gateway to
     assign frames to a Gateway Conversation ID. 8.2 provides
     the IEEE 802.1 OUI (00-80-C2) Gateway Algorithm
     encodings.  Default is the value for an unspecified
     distribution algorithm.";
  reference
    "7.4.1.1.6 of IEEE Std 802.1AX";
}
container home-gateway-conv-service-map {
  if-feature "ax:sid-map";
  description
    "Data structure to map ~~service identifiers~~ Service Identifiers to
     ~~conversation identifiers~~Conversation Identifiers. Each entry consists of a
     Conversation ID (CID) and a list of zero or more Service
     Identifiers (SIDs) that map to it. Frames with Service
     IDs not contained in the map are not mapped to any
     Gateway Conversation ID and are discarded.";
  reference
    "7.4.1.1.20, 6.6.3.1 of IEEE Std 802.1AX";
  uses ax:service-map;
}
leaf home-gateway-conv-service-digest {
  type binary;
  config false;
  description
```

```
    "The MD5 Digest of the home-gateway-conv-service-map. The
     value is NULL (AAAAAAAAAAAAAAAAAAAAAA== in base64) when
     the distribution algorithm specified
     by agg-port-algorithm does not use the
     home-gateway-conv-service-map.";
  reference
    "7.4.1.1.21 of IEEE Std 802.1AX";
}
container gateway-enable-mask {
  description
    "A vector of Boolean values, with one value for each
     possible Gateway Conversation ID. A 1 indicates that
     frames associated with that Gateway Conversation ID
     are allowed to pass through this Gateway Port, and a
     0 indicates that such frames are not allowed to pass.
     Default value is all bits set to 1.";
  reference
    "7.4.1.1.18, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
  uses drni-mask;
}
container gateway-preference-mask {
  description
    "A vector of Boolean values, with one value for each
     possible Gateway Conversation ID. A 1 indicates that
     this Gateway Port is the preferred Gateway when both
     DRNI Gateways have the Gateway Conversation ID enabled
     in the gateway-available-mask, and a 0 indicates that
     it is not preferred.
     Default value is all bits set to 1.";
  reference
    "7.4.1.1.19, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
  uses drni-mask;
}
container gateway-available-mask {
  config false;
  description
    "A vector of Boolean values, with one value for each
     possible Gateway Conversation ID. A 1 indicates that
     this Gateway Port is eligible to be selected to pass
     that Gateway Conversation ID, and a 0 indicates that
     it is not eligible.";
  reference
    "7.4.1.1.22, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
  uses drni-mask;
}
container neighbor {
  config false;
  description
    "Operational values for the DRNI neighbor obtained
     from DRCPDUs.";
  leaf oper-irp-state {
    type irp-state;
    description
      "A string of 8 bits, corresponding to the current
       operational value of IRP_State as transmitted in
       DRCPDUs.";
    reference
      "7.4.1.1.26, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
  }
  leaf system {
    type ieee:mac-address;
    description
      "The MAC Address portion of the System Identifier of
       the Neighbor DRNI System (connected via the
       Intra-Relay Port). ";
    reference
      "7.4.1.1.29 of IEEE Std 802.1AX";
  }
  leaf system-priority {
    type uint16;
    description
      "The priority portion of the System Identifier of the
```

```
     Neighbor DRNI System (connected via the Intra-Relay
     Port).";
  reference
    "7.4.1.1.30 of IEEE Std 802.1AX";
}
leaf drni-key {
  type uint16;
  description
    "The DRNI key value received from the Neighbor DRNI
     System (connected via the ~~IntraRelay ~~Intra-Relay Port).";
  reference
    "7.4.1.1.31 of IEEE Std 802.1AX";
}
leaf aggregator-algorithm {
  type identityref {
    base ax:distribution-algorithm;
  }
  description
    "The Port algorithm used by the Neighbor Aggregator to
     assign frames to Port Conversation IDs.";
  reference
    "7.4.1.1.33 of IEEE Std 802.1AX";
}
leaf aggregator-conv-service-digest {
  type binary;
  description
    "The MD5 Digest of the Neighbor Aggregator's
     Admin_Conv_Service_Map. Obtained from the Home
     Aggregator State TLV last received from the Neighbor
     DRNI System.";
  reference
    "7.4.1.1.34 of IEEE Std 802.1AX";
}
leaf aggregator-conv-link-digest {
  type binary;
  description
    "The MD5 Digest of the Neighbor Aggregator's
     Admin_Conv_Link_Map. Obtained from the Home Aggregator
     State TLV (9.6.2.4) last received from the Neighbor
     DRNI System.";
  reference
    "7.4.1.1.35 of IEEE Std 802.1AX";
}
leaf partner-system-priority {
  type uint16;
  description
    "The priority portion of the System Identifier of the
     Neighbor Aggregator's Partner.";
  reference
    "7.4.1.1.36 of IEEE Std 802.1AX";
}
leaf partner-system {
  type ieee:mac-address;
  description
    "The MAC Address portion of the System Identifier of
     the Neighbor Aggregator's Partner.";
  reference
    "7.4.1.1.37 of IEEE Std 802.1AX";
}
leaf partner-aggregator-key {
  type uint16;
  description
    "The operational key value of the Neighbor
     Aggregator's Partner.";
  reference
    "7.4.1.1.38 of IEEE Std 802.1AX";
}
leaf cscd-state {
  type bits {
    bit reserved-1 {
      position 0;
      description
```

```
        "Bit 1 is reserved for future use. It is set to 0
         and ignored on receipt.";
    }
    bit reserved-2 {
      position 1;
      description
        "Bit 2 is reserved for future use. It is set to 0
         and ignored on receipt.";
    }
    bit reserved-3 {
      position 2;
      description
        "Bit 3 is reserved for future use. It is set to 0
         and ignored on receipt.";
    }
    bit cscd_gateway_control {
      position 3;
      description
        "CSCD_Gateway_Control is encoded in bit 4. When
         this flag is TRUE, the DRNI Gateway is configured
         to minimize forwarding data frames on the IRC by
         selecting the DRNI Gateway and Aggregator Ports
         for forwarding any given Conversation ID to be in
         the same DRNI System.";
    }
    bit discard_wrong_conversation {
      position 4;
      description
        "Discard_Wrong_Conversation is encoded in bit 5.
         The Aggregator's Discard_Wrong_Conversation
         value.";
    }
    bit differ_conv_link_digests {
      position 5;
      description
        "Differ_Conv_Link_Digests is encoded in bit 6.
         This flag is TRUE when the Aggregator's
         Actor_Conv_Link_Digest matches the Aggregator's
         Partner_Conv_Link_Digest.";
    }
    bit differ_conv_service_digests {
      position 6;
      description
        "Differ_Conv_Service_Digests is encoded in bit 7.
         This flag is TRUE when the Aggregator's
         Actor_Conv_Service_Digest matches the Aggregator's
         Partner_Conv_Service_Digest.";
    }
    bit differ_port_algorithms {
      position 7;
      description
        "Differ_Port_Algorithms is encoded in bit 8. The
         Aggregator's differPortAlgorithms flag is TRUE
         when the Aggregator's Actor_Port_Algorithm matches
         the Aggregator's Partner_Port_Algorithm.";
    }
  }
  description
    "8 bits, corresponding to the Aggregator_CSCD_State
     in the Neighbor_Aggregator_State variable. The first
     three bits (the least significant bits of CSCD_State)
     are reserved; the fourth bit corresponds to the
     Neighbor's value for Home_Admin_CSCD_Gateway_Control;
     the fifth bit corresponds to the Neighbor Aggregator's
     operational value for Discard_Wrong_Conversation; and
     the sixth, seventh, and eighth bits correspond to the
     Neighbor Aggregator's operational value for
     differConvLinkDigests, differConvServiceDigests, and
     differPortAlgorithms, respectively, (the most
     significant bits of CSCD_State).";
  reference
    "7.4.1.1.39 of IEEE Std 802.1AX";
```

```
      }
      leaf-list active-links {
        type uint16;
        description
          "A list of the operational Link_Numbers of Aggregation
           Ports that are currently active (i.e., collecting) on
           the Neighbor's Aggregator. An empty list indicates that
           there are no Aggregation Ports active. Each integer
           value in the list carries an aAggPortOperLinkNumber
           attribute value.";
        reference
          "7.4.1.1.40 of IEEE Std 802.1AX";
      }
      leaf gateway-algorithm {
        type identityref {
          base ax:distribution-algorithm;
        }
        description
          "The gateway algorithm used by the Neighbor DRNI
           Gateway to assign frames to Gateway Conversation IDs.";
        reference
          "7.4.1.1.41 of IEEE Std 802.1AX";
      }
      leaf gateway-conv-service-digest {
        type binary;
        description
          "The MD5 Digest of the Neighbor DRNI Gateway's
           the Home_Admin_Gateway_Conv_Service_Map. Obtained
           from Gateway_Conv_Service_Digest in the
           Neighbor_Gateway_State TLV last received from the
           Neighbor DRNI System.";
        reference
          "7.4.1.1.42 of IEEE Std 802.1AX";
      }
      container gateway-available-mask {
        description
          "A vector of Boolean values, with one value for each
           possible Gateway Conversation ID. A 1 indicates that
           the Neighbor DRNI Gateway Port is eligible to be
           selected to pass that Gateway Conversation ID, and
           a 0 indicates that it is not eligible.";
        reference
          "7.4.1.1.43 of IEEE Std 802.1AX";
        uses drni-mask;
      }
      container gateway-preference-mask {
        description
          "A vector of Boolean values, with one value for each
           possible Gateway Conversation ID. A 1 indicates that
           the Neighbor DRNI Gateway Port is the preferred
           Gateway when both DRNI Gateways have the Gateway
           Conversation ID enabled in the gateway-available-mask,
           and a 0 indicates that it is not preferred.";
        reference
          "7.4.1.1.44 of IEEE Std 802.1AX";
        uses drni-mask;
      }
    }
  }
}

augment "/if:interfaces/if:interface/if:statistics"
      + "/dot1ax:lag-stats" {
  when '../../dot1ax:lag/dot1ax-drni:drni' {
    description
      "Applies to aggregators Aggregators with DRNI present.";
  }
  description
    "Augment interface statistics with DRNI statistics.";
  container drni-stats {
    description
      "Contains DRNI specific statistics.";
```

```
      leaf drcpdus-rx {
        type yang:counter64;
        description
          "The number of valid DRCPDUs received on this
           Intra-Relay Port.";
        reference
          "7.4.1.1.45 of IEEE Std 802.1AX";
      }
      leaf illegal-rx {
        type yang:counter64;
        description
          "The number of frames received on this Intra-Relay
           Port that carry the DRCP EtherType value,
           but contain a badly formed PDU.";
        reference
          "7.4.1.1.46, 9.6.1.4 of IEEE Std 802.1AX";
      }
      leaf drcpdus-tx {
        type yang:counter64;
        description
          "The number of valid DRCPDUs transmitted on this
           Intra-Relay Port.";
        reference
          "7.4.1.1.47 of IEEE Std 802.1AX";
      }
    }
  }
}
```